

ORIENT AI: Intelligent Guidance

An AI-Driven Application for Predicting Academic Fit and Career Satisfaction

Tutors: **Joanna Jojczyk & Zdzisława Rowińska**
Department of Computer Science & ML Sub-Teams

| Meet the Team

A multidisciplinary engineering group split into software development and machine learning sub-teams to build Orient AI.



Aroa Aguiño
Domínguez



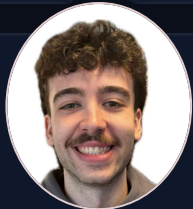
María Encina
Díez del Río



Jorge Domínguez
Palomo



Jaime Esteban
Reano



Mario Fernández
Costello



Naroa Genicio
Iglesias



David Sánchez
Sánchez

| The Vocational Crisis

Traditional systems are failing. Students face structural disorientation and friction in their careers.

- ✗ Obsolete and static instruments
- ✗ Blind elimination strategy
- ✗ Severe personal impact (dropout)



State of the Art: 2025-2026

Pymetrics (Neuroscience-Based)

Method: Uses neuroscience games and AI to assess cognitive and emotional traits.

Goal: Matches users with roles where similar profiles have succeeded.

Kickresume (AI Career Coach)

Method: Analyzes resumes using LLMs (Large Language Models).

Goal: Identifies skill gaps and suggests career paths for professional transitions.

AlCareerCoach.org (Conversational Mentor)

Method: A free, chatbot-style tool by CareerVillage.

Goal: Acts as a mentor for students to explore interests through dialogue.

Steppingblocks (Big Data Insights)

Method: Educational data analytics using "Big Data."

Goal: Shows real-world career paths based on the outcomes of millions of professionals.

IBM Watson Career Coach (Corporate NLP)

Method: Utilizes Natural Language Processing (NLP).

Goal: Helps employees find internal growth opportunities aligned with their skills.

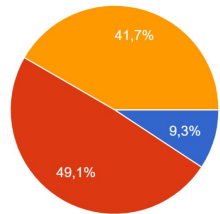
Empirical Field Validation



Qualitative Validation

Right now, do you have a clear idea of what you'll study after high school?

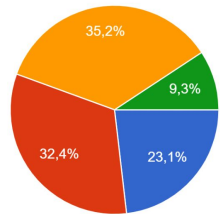
108 respuestas



- Yes, I'm 100% sure
- I have 2 or 3 options in mind, but I'm unsure
- I have absolutely no idea, I feel really lost

When you think about choosing your major or program, what is your main feeling?

108 respuestas



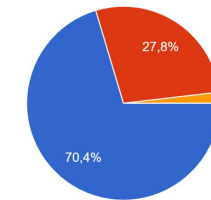
- Excited and eager to begin this new chapter
- Some uncertainty and doubts, but I'm managing for now
- Feeling a lot of pressure and afraid of choosing wrongly
- I try to avoid thinking about it so I don't stress out for now



Quantitative Survey

If there were a free mobile app that analyzed your real interests and recommended degrees that perfectly fit you, would you use it?

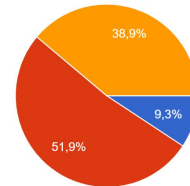
108 respuestas



- Yes, without a doubt
- Maybe, I would try it to see what it suggests
- No, I prefer to look for information on my own

Do you feel you receive enough useful, clear, and personalized information at your school about the different degrees?

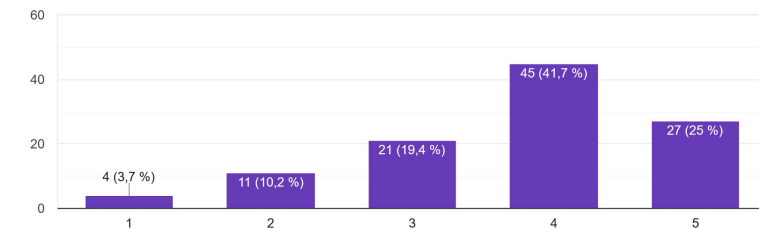
108 respuestas



- Yes, the school's guidance helps me a lot
- Partially, I get general information, but I feel something more personalized is missing
- No, the information is scarce, unclear, or outdated

From 1 (low) to 5 (high), how much stress do you feel about making this decision?

108 respuestas



OUR SOLUTION: ORIENT AI

Our intelligent framework goes beyond standard classification. By modeling psychological profiles, it bridges the gap between student aspirations and academic realities.



1. Predict "Where"



2. Predict "How Much"



3. Predict "What"

The users/stakeholders: Who We're Building For



Primary: Students



Secondary: Counselors &
Parents



Tertiary: Universities &
Employers

| Collaborative Team Architecture

To execute the project, the team divided into two specialized sub-teams with distinct technical briefs, unified through a chained JSON pipeline handoff.



Software Development Team

Streamlit / Python

UX/UI Custom CSS



Machine Learning Sub-Team

Orange Data Mining

Scikit-Learn / ML

COLLABORATIVE TEAM ARCHITECTURE

Software Sub-Team

UI/UX & INTEGRATION

- Engineered **Streamlit** web interface
- Custom Dark-Navy CSS design system
- Interactive 25-question Likert survey
- JSON handoff logic for Stage 1/2 fusion

ML Sub-Team

DATA SCIENCE & MODELING

- 145,000 OpenPsychometrics records cleaned
- Gold Dataset generation (72,000 instances)
- Chained Random Forest & Linear Regression
- Synthetic Data Augmentation (SMOTE)

SOFTWARE DEVELOPMENT



```
mirror_mod = modifier_ob.  
set mirror object to mirror  
mirror_mod.mirror_object =  
operation == "MIRROR_X":  
mirror_mod.use_x = True  
mirror_mod.use_y = False  
mirror_mod.use_z = False  
operation == "MIRROR_Y":  
mirror_mod.use_x = False  
mirror_mod.use_y = True  
mirror_mod.use_z = False  
operation == "MIRROR_Z":  
mirror_mod.use_x = False  
mirror_mod.use_y = False  
mirror_mod.use_z = True  
selection at the end -add  
mirror_ob.select= 1  
modifier_ob.select=1  
context.scene.objects.active  
("Selected" + str(modifier  
mirror_ob.select = 0  
= bpy.context.selected_obj  
data.objects[one.name].sel  
print("please select exact  
-- OPERATOR CLASSES ----  
types.Operator):  
X mirror to the selected  
object.mirror_mirror_x"  
mirror X"  
context):  
context.active_object is not
```

SELECTING THE UI FRAMEWORK

TECHNOLOGY

ADVANTAGES

DISADVANTAGES

PyQt

Native performance, rich widgets

Difficult web deployment, high UI overhead

Flask / FastAPI

Total flexibility, industry standard

Requires manual frontend development

Streamlit 

Rapid prototyping, Native Python

Limited complex state management

CONCLUSION: We selected **Streamlit** for its seamless integration with the Python ML ecosystem and speed of delivery.

| SOFTWARE REQUIREMENTS



ACCESSIBILITY

**Universal
access across
browsers.**



SCALABILITY

**Support for
growing user
loads.**



MAINTAINABILITY

**Modular code
architecture.**



AI INTEGRATION

**Frictionless
ML data flow.**



USER EXPERIENCE

**Professional
startup feel.**

SYSTEM ARCHITECTURE



| WHY A WEB APPLICATION?

DESKTOP APP

- ✗ Installation required
- ✗ OS-specific builds
- ✗ Friction-heavy updates

WEB APPLICATION

- ✓ Zero installation
- ✓ Cross-platform access
- ✓ Easy cloud deployment
- ✓ Instant scaling

| MODULAR DESIGN PHILOSOPHY



SEPARATION OF CONCERNS

Each module performs one task, and one task only.



INDEPENDENT TESTING

Debug the UI without affecting the ML logic.



FUTURE SCALABILITY

Easily swap modules for new features or AI models.

"A modular foundation allows ORIENT AI to evolve from a project to a scalable product."

USER DATA STRUCTURE

Raw input is transformed into a clean, multi-dimensional feature vector.

THE INPUT MODEL



Demographics

Gender identity



Psychometrics

25 Likert questions



Ground Truth

Satisfaction Score (0–100)

The architecture ensures that **heterogeneous raw data** is normalized through a validation engine before being fed into the primary machine learning pipeline, maintaining **100%** data fidelity.

| VALIDATION STRATEGY



INTEGRITY CHECKS

Preventing missing answers or null values from reaching the model.



TYPE ENFORCEMENT

Enforcing strict data formats (Float64) for mathematical consistency.

RELIABLE OUTPUT

Ensuring 100% data fidelity for AI-ready inputs.

Raw Input



VALIDATION ENGINE



GOLD DATASET

DATA HANDOFF MECHANISM

The JSON Bridge

We engineered a custom handoff mechanism that bridges the gap between software engineering and data science.



Standardized Payload

Consistent JSON schema for reliable data ingestion.



Interoperability

Seamless integration across diverse Python libraries.



Scalability

Architected for high-speed real-time inference.



PREPARING FOR AI INTEGRATION

COLLECT > VALIDATE > STRUCTURE > TRANSFER

"The Software Team has transformed student responses into reliable,
AI-ready data."

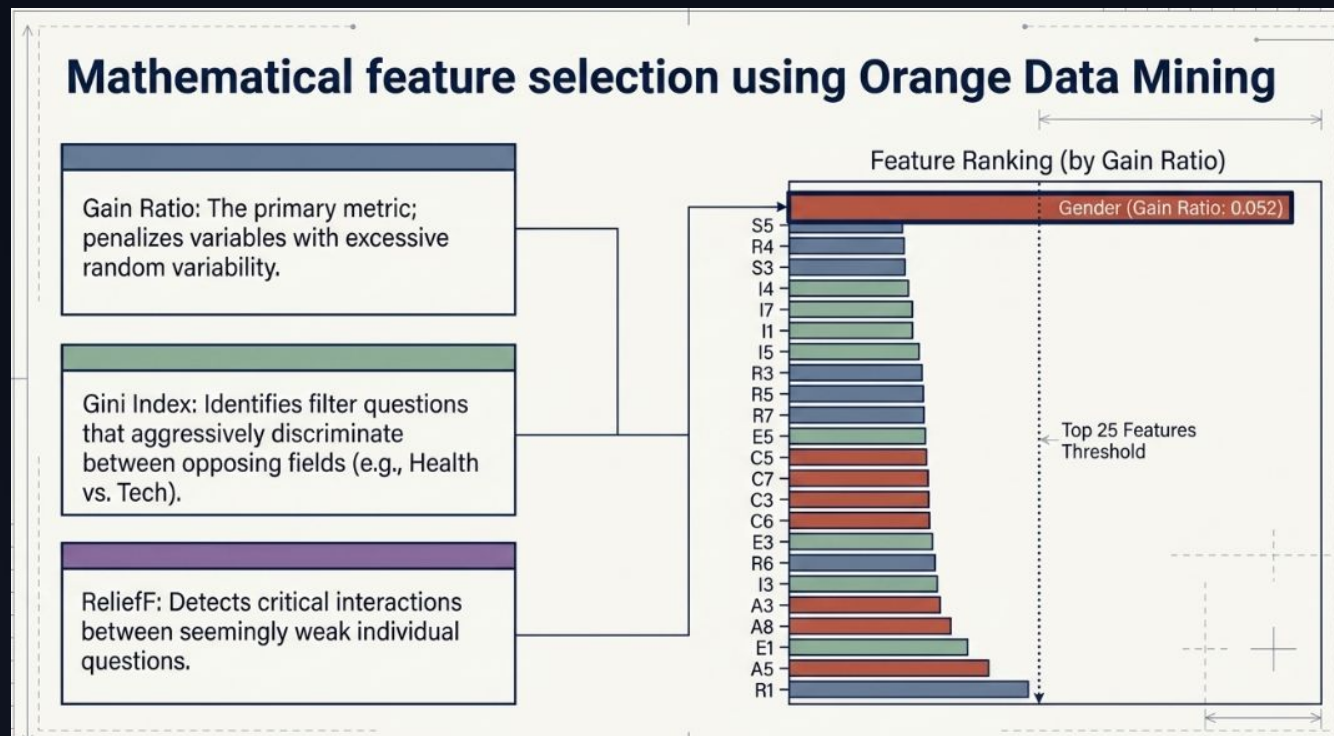
MACHINE LEARNING MODEL DEVELOPME NT



Questionnaires Generation Process

First Questionnaire: Optimization

Reference Dataset: Holland Code (RIASEC) from Open Psychometrics



		#	Gain ratio	Gini	ReliefF
1	N	gender	0.052	0.015	0.000
2	N	S5	0.032	0.019	0.021
3	N	R4	0.030	0.017	0.006
4	N	S3	0.029	0.019	0.014
5	N	I4	0.028	0.014	0.001
6	N	I7	0.027	0.013	0.010
7	N	I1	0.026	0.014	0.004
8	N	I5	0.025	0.012	0.013
9	N	R3	0.024	0.010	0.009
10	N	R5	0.021	0.009	-0.000
11	N	R7	0.019	0.011	0.003
12	N	E5	0.018	0.011	0.002
13	N	C5	0.016	0.007	-0.001
14	N	C7	0.015	0.009	0.012
15	N	C3	0.015	0.009	0.004
16	N	C6	0.014	0.007	0.016
17	N	E3	0.014	0.009	0.010
18	N	R6	0.013	0.008	0.003
19	N	I3	0.013	0.007	-0.003
20	N	A3	0.013	0.003	0.000
21	N	A8	0.013	0.003	0.017
22	N	E1	0.012	0.007	0.005
23	N	A5	0.012	0.003	0.016
24	N	R1	0.011	0.006	0.012

| Questionnaires Generation Process

Second Questionnaire: Implementation

Main Objectives:

- Mapping of the Existing Academic Offer in Łódź
- Strategic Creation of Personalized Questions

Machine Needed

Learning

Models

We developed three separate machine learning models for the app. Each model focusing on a single target variable.

Model 1

Questionnaire 1

Target: area of study

Model 2

Questionnaire 1

Target: satisfaction

Model 3

Questionnaire 2

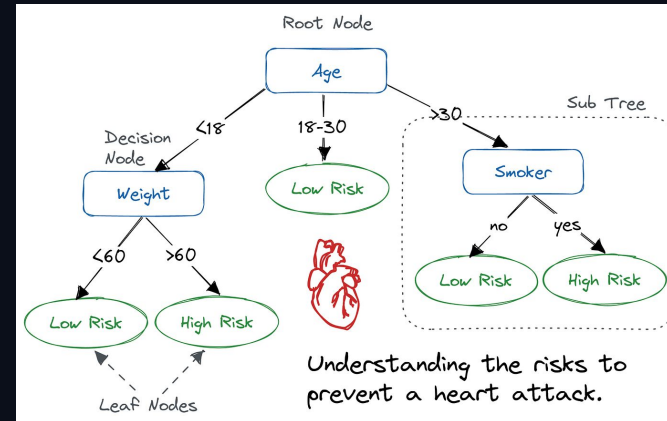
Target: career of
choice

We need 3 models Selected on synthetic data

Explanation of the Tested Models

CLASSIFICATION

- Logistic Regression
- Naive Bayes
- Decision Tree
- KNN
- SVM
- Random Forest

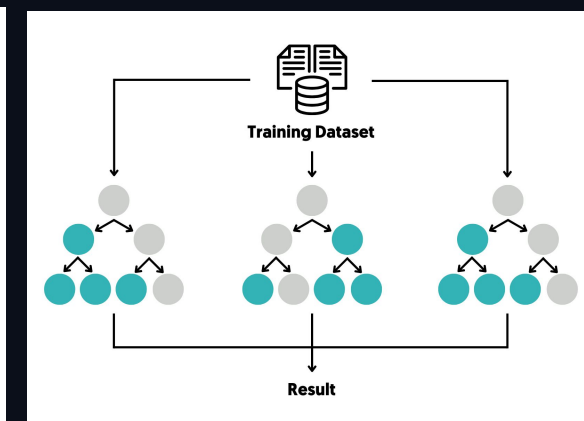
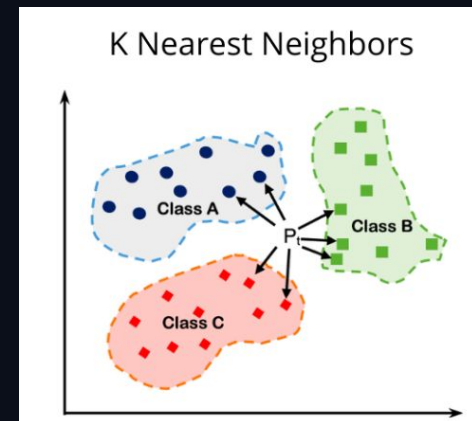
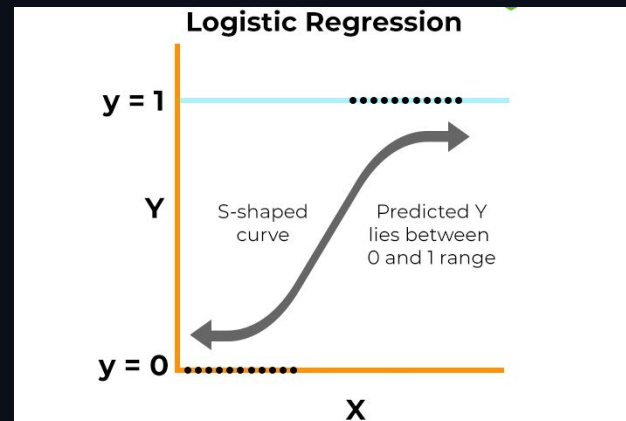
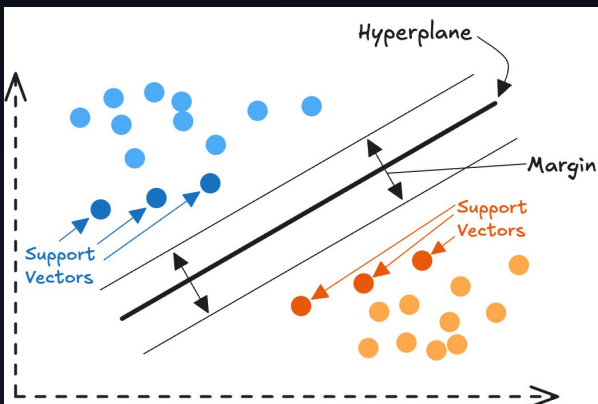


Likelihood Class Prior Probability

$$P(c|x) = \frac{P(x|c)P(c)}{P(x)}$$

Posterior Probability Predictor Prior Probability

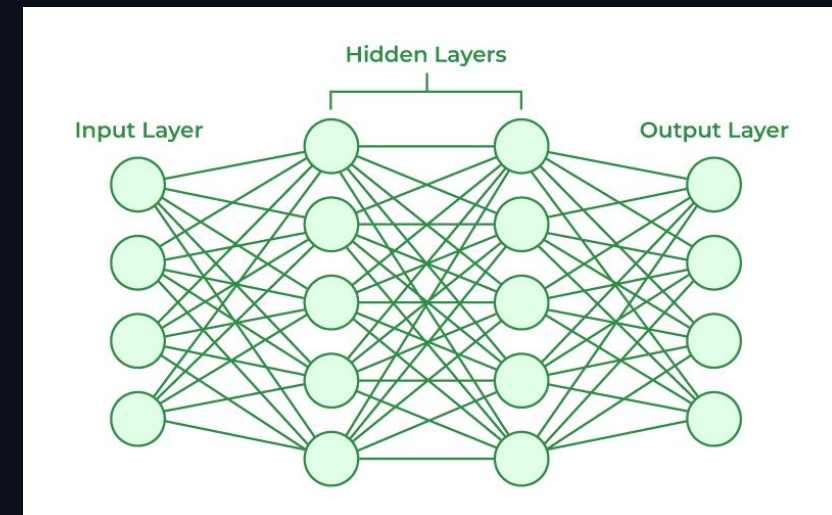
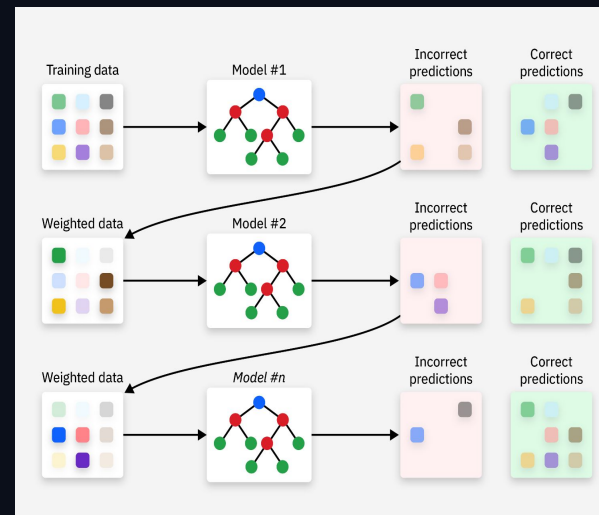
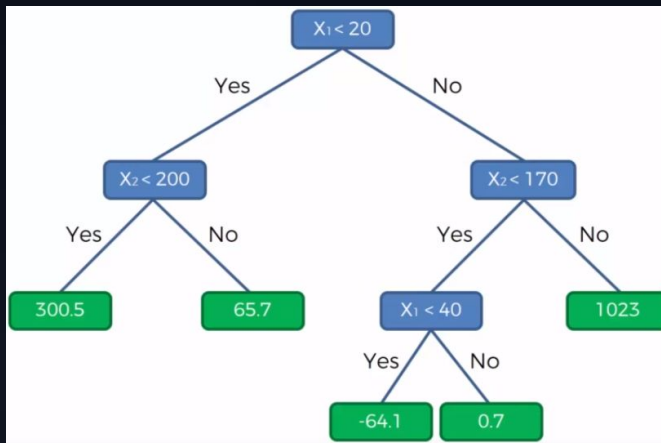
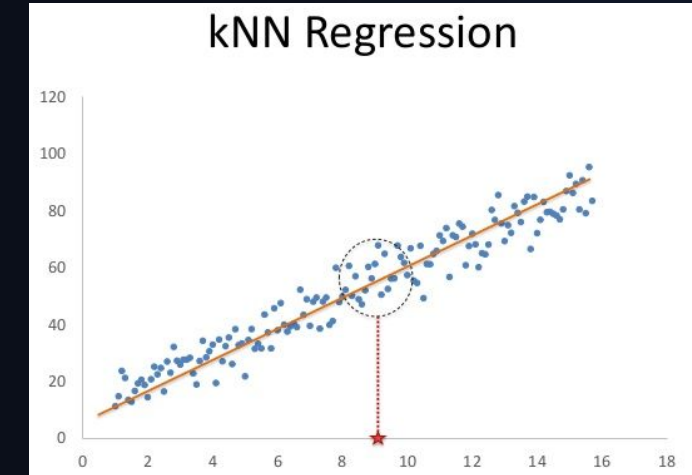
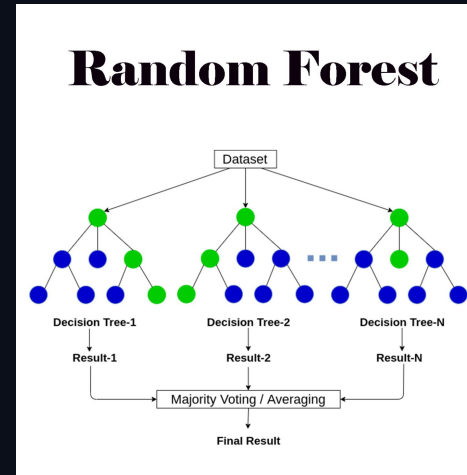
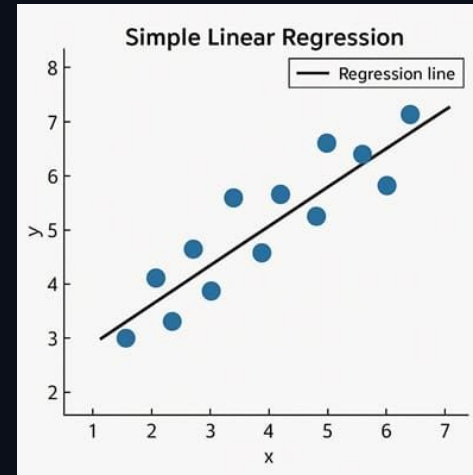
$$P(c|X) = P(x_1|c) \times P(x_2|c) \times \dots \times P(x_n|c) \times P(c)$$



Explanation of the Tested Models

REGRESSION

- Linear Regression
- Random Forest Regressor
- Gradient Boosting
- Decision Tree Regressor
- KNN Regressor
- Neural Network



Model 1: Questionnaire 1 (target-area of study)

Model	AUC	CA	F1	Prec	Recall	MCC
Logistic Regression	0.982	0.830	0.830	0.833	0.830	0.796
Naive Bayes	0.992	0.915	0.914	0.916	0.915	0.898
Tree	0.820	0.640	0.644	0.652	0.640	0.568
kNN	0.905	0.650	0.644	0.649	0.650	0.580
SVM	0.978	0.805	0.804	0.809	0.805	0.766
Random Forest	0.943	0.780	0.776	0.782	0.780	0.736

Why Random Forest?

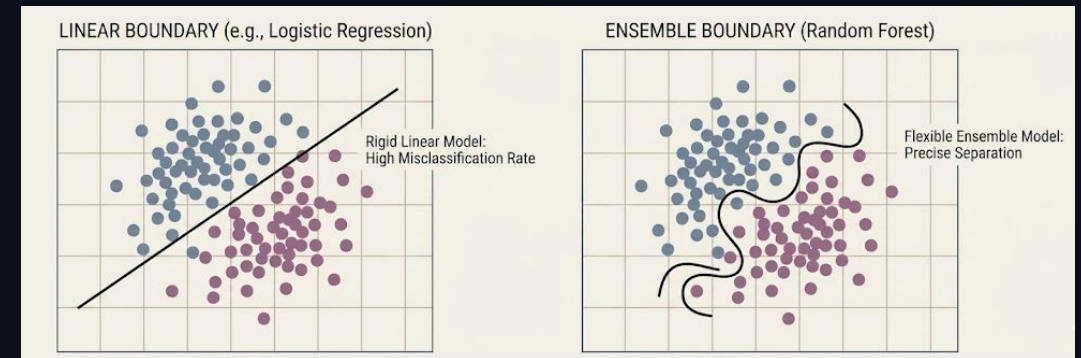
Superior handling of overlapping academic fields

High resistance to overfitting compared to single Decision Trees

unmatched robustness without requiring tedious data preprocessing

makes no assumptions and is far more resilient to noisy data

		Predicted						Σ
		arts	health	humanities	sciences	social sciences	technology	
Actual	arts	37	0	3	0	1	0	41
	health	1	12	6	1	7	0	27
	humanities	5	1	26	0	0	1	33
	sciences	0	3	1	25	2	0	31
	social sciences	2	3	0	0	26	3	34
	technology	1	0	2	0	1	30	34
Σ		46	19	38	26	37	34	200



Model 2: Questionnaire 1 (target-satisfaction)

Linear Regression: Optimized for Stability



Achieved lowest overall error metrics.
High robustness and low risk of overfitting.

Simplicity Yields Stability.

Neural Network: The Impact of Complexity



Critical Negative Fit. Advanced neural layers collapsed during training.

Complex networks overfit psychometric variables on smaller datasets, highlighting the necessity of linear boundaries.

The Pitfalls of Complexity.

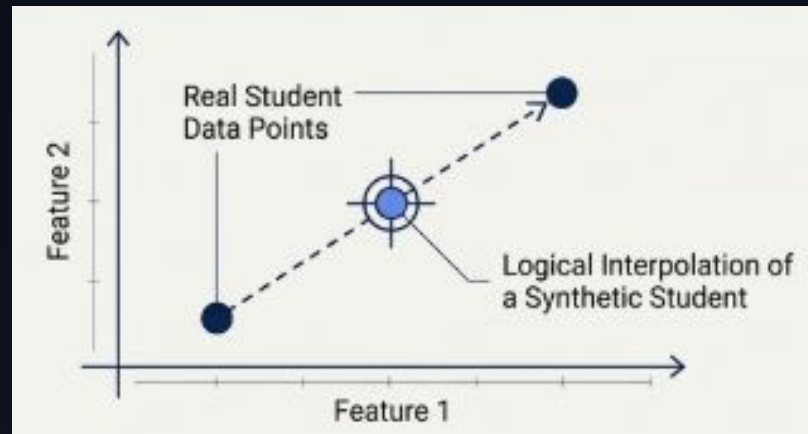
Model 3: Questionnaire 2 (target-degree)

Model	AUC	CA	F1	Prec	Recall	MCC
Random Forest	0.982	0.889	0.889	0.889	0.889	0.880
Neural Network	0.981	0.839	0.840	0.841	0.839	0.827
kNN	0.980	0.875	0.876	0.881	0.875	0.866
Gradient Boosting	0.975	0.819	0.819	0.822	0.819	0.806
Naive Bayes	0.987	0.919	0.919	0.920	0.919	0.913
Logistic Regression	0.977	0.812	0.812	0.814	0.812	0.797

DATA AUGMENTATION

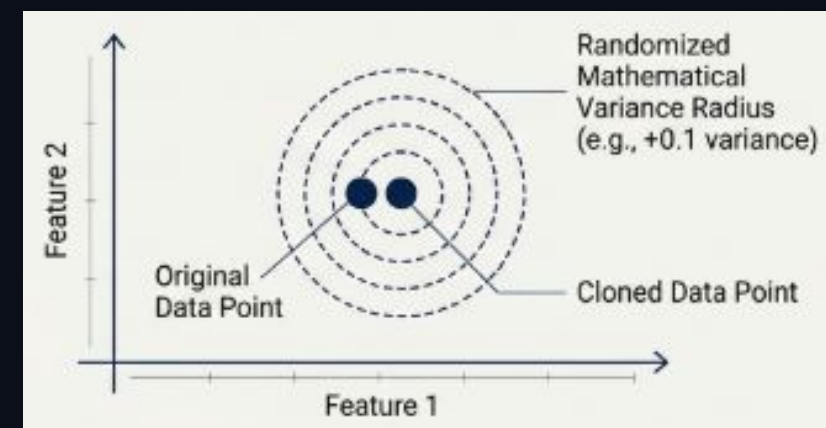
SMOTE Strategy

Synthetic Minority Over-sampling: Synthetic data is generated by interpolation between existing minority instances, filling the data space without memorization.

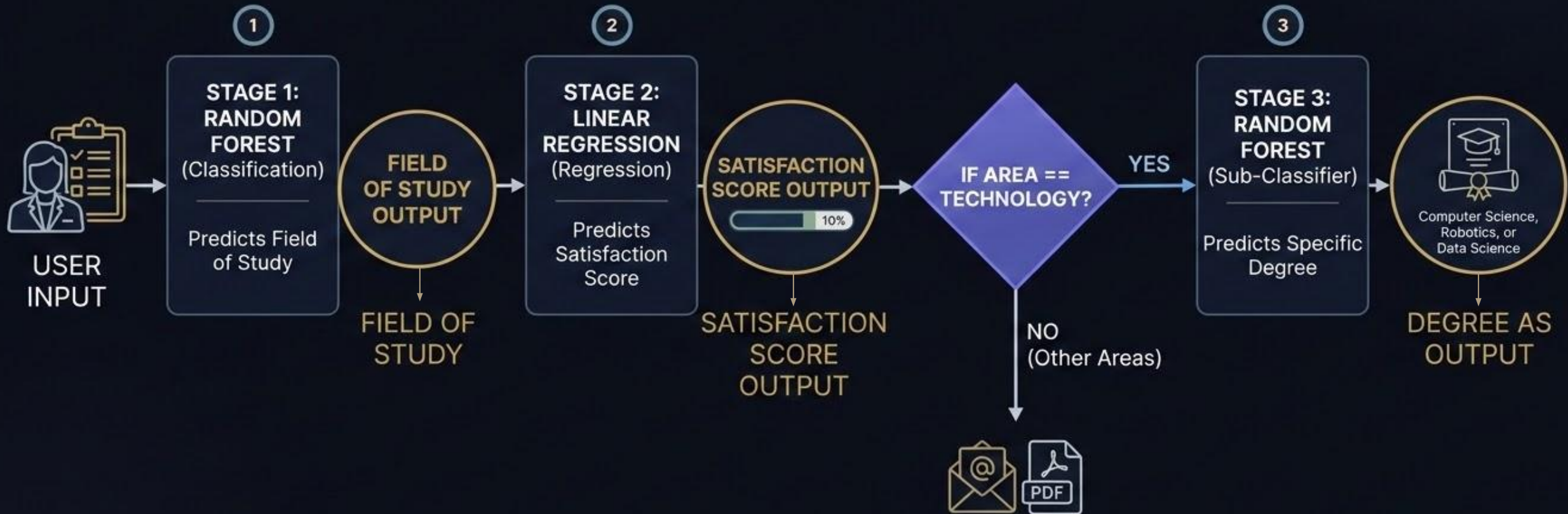


Gaussian Noise Injection

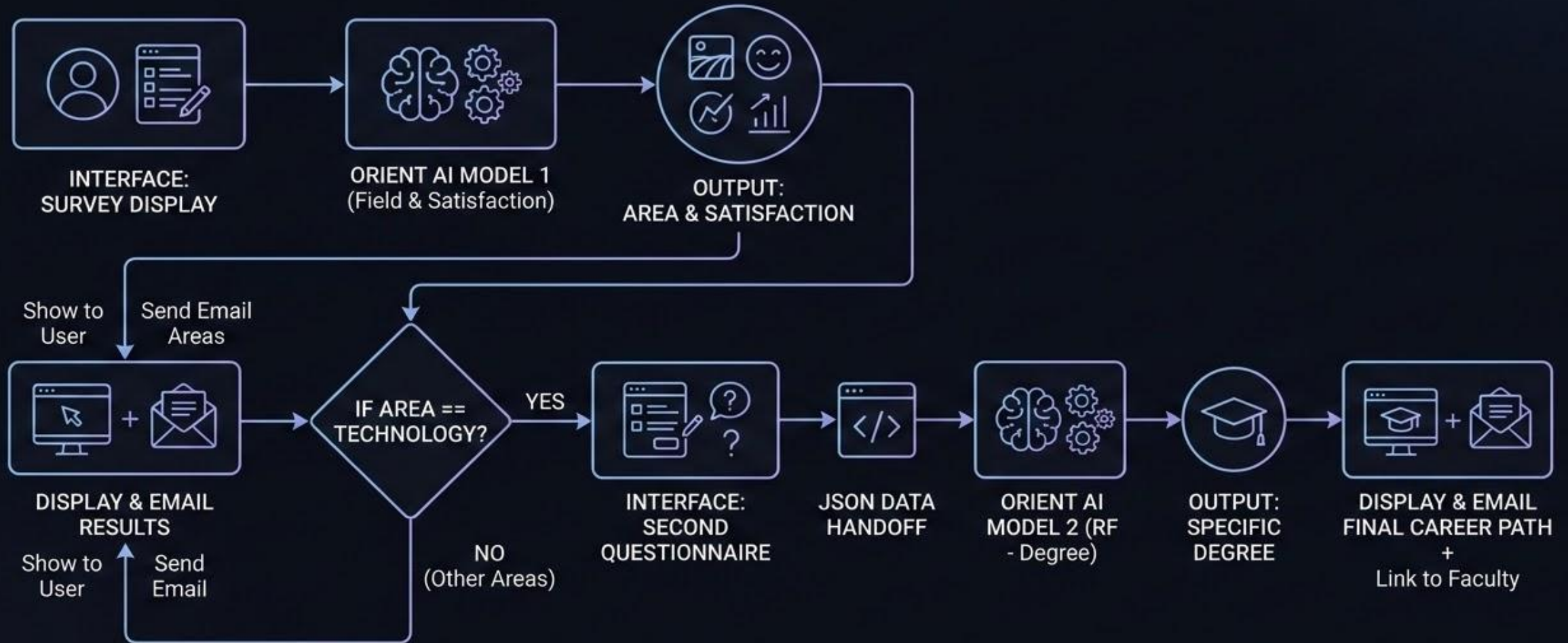
Small amounts of random Gaussian noise (± 0.1 to ± 0.2) are added to cloned minority samples to introduce variation and prevent overfitting.



The Chained Prediction Pipeline



Integration AI Model + Software



ORIENT AI



<https://orientai-deployment.streamlit.app/>



how this work?



Of course! Let me explain how OrientAI works, step by step: OrientAI is a platform designed to help you discover your ideal field of study and predict your future career satisfaction, all powered by artificial intelligence. Here's how the process unfolds: 1. **Part 1: General Orientation** * You'll begin by entering your Name, Surname, and Email (so we can send you your detailed results later), and selecting your gender. * Next, you'll answer a questionnaire consisting of 25 tasks. For each task, you'll simply rate how much you would enjoy performing it on a scale of 1 (Strongly Dislike) to 5 (Strongly Enjoy). * Based on your responses, the application will use a Random Forest model to predict your ideal **general field of study** (e.g., Humanities, Sciences, Technologies) and a Linear Regression model to predict your expected **career satisfaction score**. 2. **Part 2: Technical Specialization (if applicable)** * If your predicted general field of study from Part 1 is "Technologies" (or a related technical field), the application will unlock a second part. * This Part 2 is a more specialized survey, also with 25 questions, that focuses purely on various engineering and technical scenarios. * This section uses another Random Forest model to help you discover your more specific **engineering specialty** (e.g., Software Engineering, Mechanical Engineering, Biotechnology

ACADEMIC ORIENTATION SYSTEM

ORIENT AI

An AI-powered questionnaire that maps your interests to academic fields. Your answers will contribute to training a machine learning model that predicts the ideal university degree for your profile.

Step 1 of 27

PERSONAL DETAILS

Please enter your details to receive the survey results.

Name

Surname

Email

TASK INTEREST RATING

Rate how much you would enjoy the following activity.

Q8. Test the quality of industrial machinery or manufacturing processes.

☐ 1 — Strongly Dislike ☐ 2 — Dislike ☐ 3 — Neutral ☐ 4 — Enjoy ☒ 5 — Strongly Enjoy

★★★★★

 Back

Next 

TASK INTEREST RATING

Rate how much you would enjoy the following activity.

Q9. Conduct research on the behavior of plants or animals.

☐ 1 — Strongly Dislike ☒ 2 — Dislike ☐ 3 — Neutral ☐ 4 — Enjoy ☐ 5 — Strongly Enjoy



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AI CAREER PREDICTION

Based on your responses, your ideal field of study is: **Technologies**

Technology Profile Detected

Your interests strongly align with the Technology field. **Please complete the next specific questionnaire** to pinpoint the exact engineering degree that best suits you!

FIELD OF STUDY SATISFACTION

73/100

 Moderate Satisfaction

Field of study satisfaction



Based on academic profile analysis — how much you are likely to enjoy your chosen field.

Continue to Engineering Questionnaire 

TASK INTEREST RATING

Rate how much you would enjoy the following activity.

Q5. I'm drawn to the idea of planning supply networks (water, sanitation) that connect and sustain an urban center.

☐ 1 – Strongly Dislike ☐ 2 – Dislike ☐ 3 – Neutral ☐ 4 – Enjoy ☐ 5 – Strongly Enjoy

required

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Great job, david sanchez!

Your personalised academic orientation report is ready below.



Responses recorded successfully

Your answers will help train our ORIENT AI recommendation model. Thank you!

🌐 AI CAREER PREDICTION FOR DAVID SANCHEZ

Civil Engineering

Your ideal field of study based on your personality & interests

[TUL Civil Engineering Admission Portal ➤](#)



Results sent to your inbox!

A personalised academic report has been sent to davidxdmod@gmail.com. Please check your inbox (and spam folder) for your results.

[↩ Submit another response](#)



Great job, david sanchez!

Your personalised academic orientation report is ready below.



Responses recorded successfully

Your answers will help train our ORIENT AI recommendation model. Thank you!

⊕ AI CAREER PREDICTION FOR DAVID SANCHEZ

Arts

Your ideal field of study based on your personality & interests

Academy of Fine Arts ↗

Łódź Film School ↗

FIELD OF STUDY SATISFACTION

100/100

🌟 High Satisfaction

Field of study satisfaction

Based on academic profile analysis — how much you are likely to enjoy your chosen field.



Results sent to your inbox!

A personalised academic report has been sent to davidxdmod@gmail.com. Please check your inbox (and spam folder) for your results.

↩ Submit another response



ORIENT AI — Academic Report

Hi david sanchez,

Thank you for completing the OrientAI survey. We have analysed your responses and prepared your personalised academic orientation report below.

Field of study satisfaction

73/100

Moderate Satisfaction 😊

This score estimates how much you are likely to enjoy working in your recommended field, based on your academic profile.

 AI CAREER PREDICTION FOR DAVID SANCHEZ

Civil Engineering

Your ideal field of study based on your personality & interests

Recommended Admission & Study Program Links

Explore official degree options for your recommended profile in **Civil Engineering**:

- [TUL Civil Engineering Admission Portal](#) →

ORIENT AI — Academic Report

Hi david sanchez,

Thank you for completing the OrientAI survey. We have analysed your responses and prepared your personalised academic orientation report below.

Field of study satisfaction

100 / 100

High Satisfaction 🌟

This score estimates how much you are likely to enjoy working in your recommended field, based on your academic profile.

 AI CAREER PREDICTION FOR DAVID SANCHEZ

Arts

Your ideal field of study based on your personality & interests

Recommended Admission & Study Program Links

Explore official degree options for your recommended profile in **Arts**:

- [Academy of Fine Arts →](#)
- [Łódź Film School →](#)

...

THE FUTURE OF ORIENT AI



Educational Centers

Integration in high schools to guide students pre-selection.



Continuous Learning

Re-training based on real-time feedback loops from graduates.



Global Scalability

Deploying to AWS/Cloud to reach international markets.

Let's make academic selection objective, predictable, and reachable.

Questions & Answers



GET IN TOUCH!

Contact:

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